

Mapping Event Participation: Building an Open-Source Network Visualization Template for the Tutor/Mentor Institute

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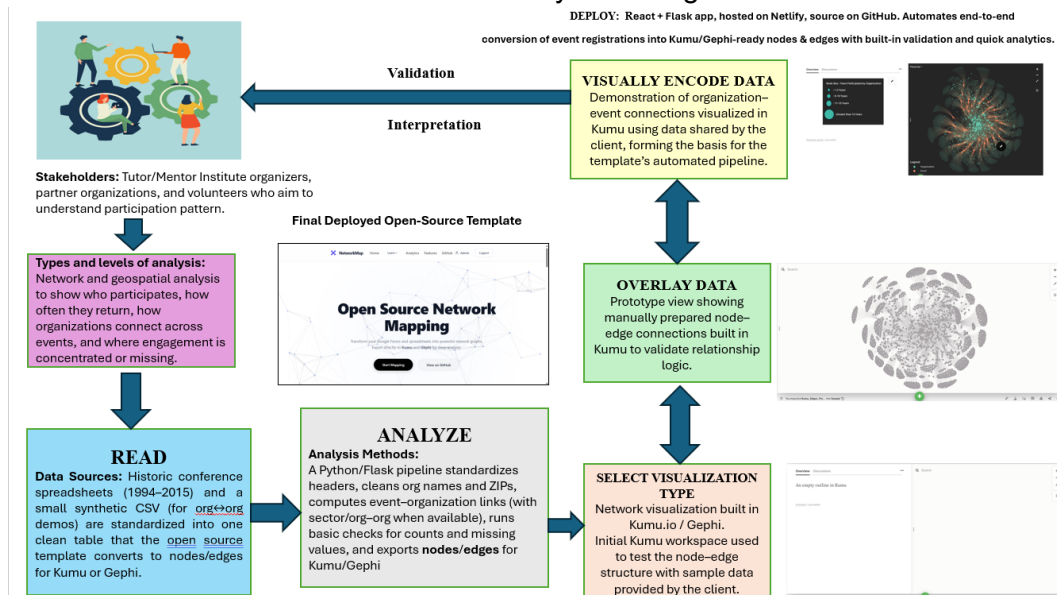


Figure 1. Visual Abstract Figure. End-to-end pipeline for Tutor/Mentor event participation; from data collection and cleaning to nodes/edges export, validation in Kumu/Gephi, and deployment of the open-source template highlighting collaboration patterns and participation gaps across years.

1. INTRODUCTION

The Tutor/Mentor Institute, LLC (formerly Tutor/Mentor Connection) is a Chicago initiative that strengthens networks for youth mentoring in underserved neighborhoods. Since 1994 it has run semi-annual Leadership & Networking Conferences, producing thousands of rows of participation data (names, organizations, roles, locations) spread across separate spreadsheets, making it hard to see who returns, where cross-sector bridges form, and who is missing.

Prior IVMOOC work and T/M blog posts [1] showed that network maps can answer these questions, but past workflows were manual and hard to repeat schemas drifted, cleaning took time, and explicit relationships were inconsistently captured. This project delivers an open-source, reusable template that standardizes registration fields and exports validated nodes and edges CSVs. Rather than build the visuals, it outputs clean, repeatable files that load directly into Kumu or Gephi. The Collect-Validate-Convert pipeline lets the Institute refresh visuals after each conference, track collaboration over time, and target outreach where engagement gaps persist.

1.1 Stakeholder Groups

Key stakeholders include: the sponsor (Daniel F. Bassill/Tutor-Mentor Institute), community organizations and educators, researchers/students, and the broader public/funders. The sponsor needs a stable open-source template; practitioners

want clear views of cross-sector connections; researchers need a clean, documented dataset; the public and funders need trustworthy visuals that communicate reach and gaps.

1.2 Stakeholder Needs

To meet these expectations, the sponsor needs a user-friendly, repeatable Collect → Validate → Convert workflow with stable schemas and IDs that reliably outputs nodes/edges CSVs and links to interactive views. Community organizations and educators need clear, filterable network visualizations by year, sector, and location that highlight anchor organizations, bridge actors, and under-represented groups so outreach can be targeted. Researchers and students need public documentation, a data dictionary, and versioned scripts that make cleaning steps transparent and fully reproducible. The public, funders, and policymakers need accessible summaries paired with high-resolution figures and interactive links, along with a brief statement of methods and caveats to support confident decision-making.

2. DATA ACQUISITION

We obtained the historical Tutor/Mentor conference records directly from the sponsor via a shared Google Drive. Each event (1994–2015) is stored as a separate .xls file with attendee details such as name, organization, role/title, sector (SNA

code), city, state, ZIP, and contact fields. We pulled all yearly files into one workspace, documented header differences, and set up a repeatable path from raw sheets to a single cleaned table that we can hand off to the export step.

2.1 Data Sources

The primary sources are 40+ Excel spreadsheets named by event and year (e.g., 95-Nov95conf.xls, 04-Nov2004conf.xls, 13-June13Conf.xls, 14-May14Conf.xls). Sponsor webpages [2,3] and blog posts were used only to confirm event timing and the intent of certain fields. No third-party datasets were added.

2.2 Data Description, Quality and Coverage

After we merged the files, the final table has about 14,968 participant rows from about 2,671 organizations. Most entries are from the Chicago area, with more from other U.S. cities. Since the files were created separately over many years, the headers and spellings didn't always match and some fields were missing. We fixed this by making column names consistent, standardizing SNA categories and 5-digit ZIPs, making organization names consistent (using a small lookup list for common variations), removing duplicate rows, and filling or flagging missing values when we could (for example, getting the city from the ZIP code). The result is one clean table, with each row tied to its event year, ready to export as validated nodes and edges CSVs for Kumu or Gephi.

3. DATA ANALYSIS

Historic conference spreadsheets were processed in Python (pandas/numpy), with quick spot-checks in matplotlib. The workflow is Collect-Validate-Convert: load the merged person-event table (year, organization, SNA sector, title, location), standardize headers, clean organization names and SNA codes, enforce 5-digit ZIPs, and remove duplicates. The cleaned table is then exported as elements (nodes) and connections (edges) for Kumu (from/to) and Gephi (source/target), creating a reproducible bipartite network that can be regenerated as new events are added.

Two small summaries provide context for the visuals. [Fig 2] ranks cities by participation: Chicago clearly leads, with smaller clusters in neighboring cities useful signals for outreach. [Fig 3] shows SNA categories: Program organizations are most common, followed by Resource and College, reflecting a mix of youth-serving programs and supporting institutions. Together, these checks confirm the data behave as expected and set up the network views and open-source export template.

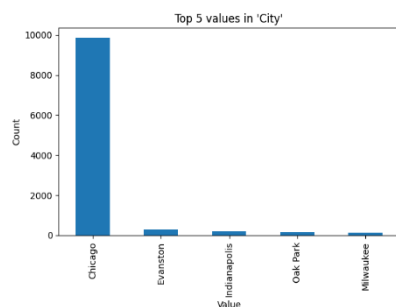


Figure 2. Top Cities (Descriptive EDA): Top-5 bar Top 5 of participant counts by city, with Chicago leading by a wide margin.

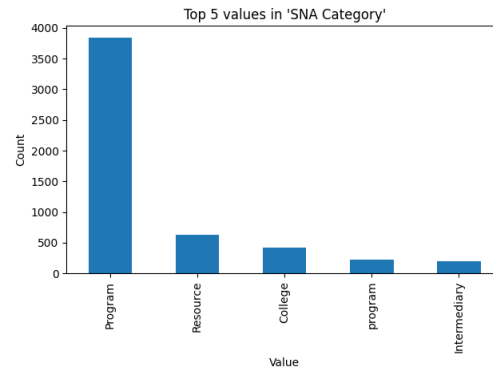


Figure 3. SNA Category (Descriptive EDA): Top-5 bar chart of SNA counts showing “Program” as the dominant category

4. EVENT NETWORK TEMPLATE

The project now delivers a working Event Network Template [7] that converts a cleaned participation table into validated Nodes and Edges CSVs for Kumu and Gephi. The web app guides users from upload to export: a landing view [Fig 4] sets the goal (open-source network mapping from forms or spreadsheets); a workflow view [Fig 5] explains the three steps, Collect-Process-Analyze/Export and the Upload & Convert screen lets users choose a graph model (e.g., Organizations ↔ Events, Organizations ↔ Organizations, or sector-based three-layer modes). After a quick schema check, the backend standardizes values (5-digit ZIPs, sector labels, common org aliases), removes duplicates, and builds relationship lists. An Analytics dashboard [Fig 6] provides sanity checks (participant/organization counts, average connections, sector/state bars). The Convert to Network Files [Fig 7] panel then reports result sizes and provides Download Nodes CSV and Download Edges CSV links (example: 87 nodes, 522 edges from a synthesized test file).

Implementation uses a React front end, Flask (Python) services for validation and graph generation, and Docker for reproducible builds. The app is deployed on Netlify, and per the client's open-source requirement the full source, issues, and setup instructions are published in a public GitHub repository. Testing to date uses synthetic datasets that follow the sponsor's standard columns; historical conference files can be processed by applying the same normalization rules at ingest.

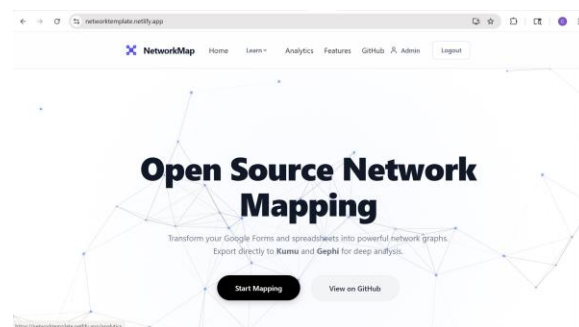


Figure 4. NetworkMap landing page; open-source tool to turn forms/spreadsheets into network graphs with Kumu/Gephi export.

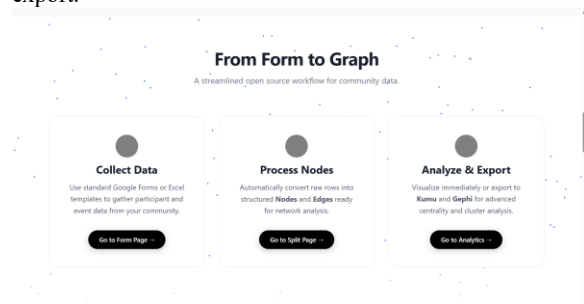


Figure 5. Workflow cards: End-to-end flow: Collect Data → Process Nodes → Analyze & Export.

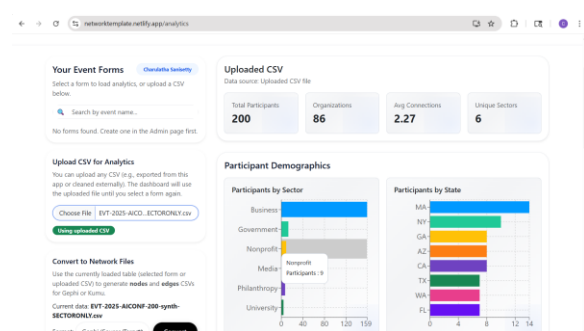


Figure 6. Analytics dashboard: Uploaded-CSV analytics showing totals, organization count, average connections, and demographic charts.

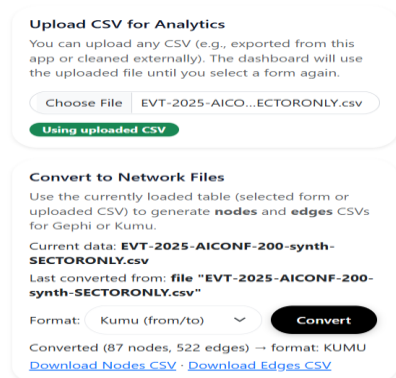


Figure 7. Convert panel: One-click generation of Kumu/Gephi-ready nodes and edges CSVs with direct download links.

5. VISUALIZATIONS

The visualizations were created in Kumu to make participation patterns easy to read and simple to refresh from the exported nodes/edges files produced by the open-source template. Two complementary views are presented from the historical dataset, a bipartite organization–event network and a U.S. ZIP heatmap and a third view demonstrates what becomes possible with the new template and enriched fields.

5.1 Organization-Event Network Visualization

This bipartite view [Fig 8] and [Fig 9].[8] places organizations (teal) and conference events (orange) in one graph and draws an edge for each attendance. Organization node size encodes years of participation, so larger teal nodes mark long-term, repeat attendees while smaller nodes indicate one-time or occasional presence. Because the historic files lacked reliable org↔org connection fields, this view focuses on organization ↔ event relationships and “years participated.” By centering any organization, the graph shows exactly which events it attended and when, surfacing enduring partners, bridge builders, and candidates for re-engagement. In short, the view answers: Which organizations return most often? Which events drew the broadest mix? Where did participation drop off?

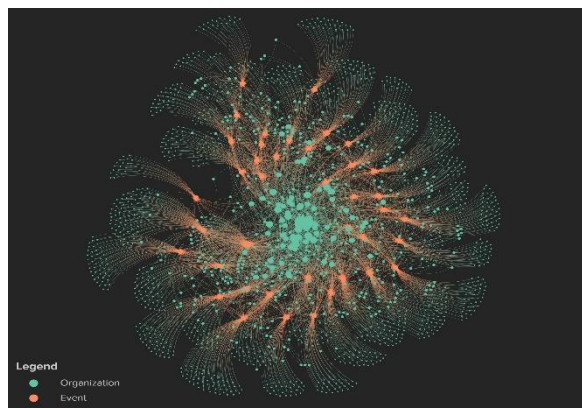


Figure 8. Kumu bipartite network of organizations and events links show attendance relationships across years.

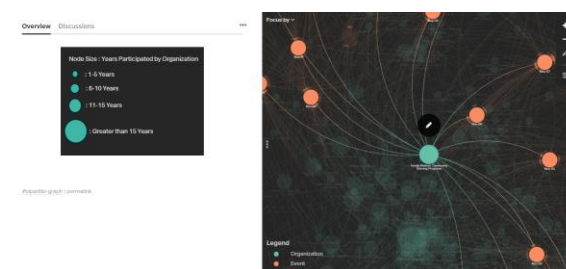


Figure 9. Kumu focus view; node size encodes years of participation; zoomed on a single organization and its linked events.

5.2 Geographic Distribution Visualization

A ZIP-level heatmap [Fig 10] shows where participants are located across the United States. Chicago appears as the most intense cluster, reflecting the program’s core focus, with additional clusters around Los Angeles, Phoenix, New York, and Washington, D.C. The map does more than show counts: it reveals coverage and gaps at a glance, highlights adjacent metros where outreach could grow, and provides a baseline to track geographic reach as new events are added. When combined with year filters, the heatmap also helps separate one-off spikes from sustained presence in a region, guiding decisions on venue, partner recruitment, and targeted

invitations

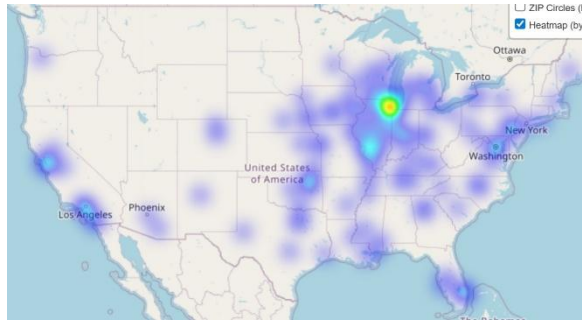


Figure 10. Heatmap showing the geographic distribution of Tutor/Mentor Conference participants by ZIP code. Warmer colors indicate higher concentrations, with Chicago emerging as the network's central hub.

5.3 Cross-sector demonstration with synthesized data

To illustrate the full capability of the template beyond what the historic files support, a third Kumu view [Fig 11] [9] was built from a synthesized dataset generated with the open-source tool. Using the standardized columns gathered per the sponsor's guidance (e.g., Organization, Event, Sector, Location, and optional org↔org ties such as "collaborated," "co-hosted," or "member-of"), the template produced clean nodes and edges that render a three-layer structure – Event to Sector to Organization, alongside organization-to-organization connections. This demonstration shows how future conferences can capture a few extra fields at registration and immediately see cross-sector tracks, intra-sector clusters, and direct org relationships that were not recoverable from the legacy spreadsheets. Practically, it answers: Which sectors each event engaged? Which organizations bridge sectors? Where do org-to-org ties suggest momentum for partnerships or sponsorships?

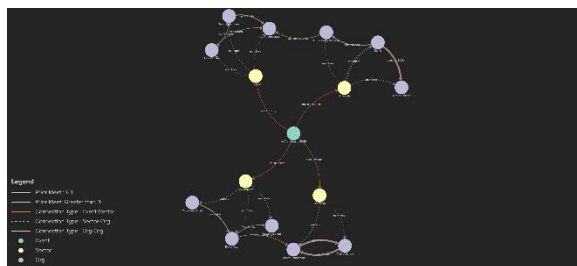


Figure 11. Kumu three-layer network from the template's synthesized data; Event–Sector–Organization and org↔org ties, showing cross-sector tracks, bridging organizations, and direct collaborations.

6. USAGE AND CRITIQUE OF AI TOOLS

AI was used as a helper for routine work, drafting starter Pandas code, suggesting targeted clean-ups (e.g., trimming ZIPs to 5 digits, standardizing SNA labels), and generating synthetic test rows so the upload-validate-export flow could be exercised end-to-end. It also offered UI suggestions for the open-source template such as clearer field hints, a compact validation panel, and more prominent download buttons, which were adopted after quick reviews. AI sped up drafting but was never treated as a source of truth. Large models sometimes

invented fields or gave inconsistent suggestions, which can reduce trust and reproducibility. To keep results reliable, all AI ideas were translated into explicit rules (header map, ZIP regex, SNA dictionary, org-name alias list), with changes reviewed by a person before they were kept.

7. INTERPRETATION OF RESULTS

Because the historic spreadsheets did not capture explicit connection fields (e.g., org↔org ties collected at registration), only one primary perspective was feasible from that archive: an organization–event network. Even with this constraint, the picture is clear, a Chicago-anchored core of repeat participants surrounded by many one-time attendees, and smaller regional clusters that point to outreach opportunities. To move beyond this single view, the open-source template now covers the full path from data collection to ready-to-load nodes and edges files. With standardized fields (Name, Organization, Sector, Location, Event/Year and optional org↔org relationships), the same pipeline can generate multiple complementary perspectives: organization–event over time, sector bridges, peer organization networks, and geography. The code and setup are published on GitHub[7], making the process reproducible for future conferences and enabling richer visualizations as more complete connection data are collected.

8. ACKNOWLEDGMENT

Grateful thanks to Daniel F. Bassill (Tutor/Mentor Institute, LLC) for the project vision, historic conference data, and continuous feedback. Appreciation to the course instructors and teaching staff for guidance and review, and to classmates for constructive peer comments. The work also benefited from the tools and platforms used throughout, Kumu, React, Flask, Python/Pandas, Docker, Netlify, and GitHub and from context provided by the Tutor/Mentor Institute website and blog, which are acknowledged as core sources for prior work and motivation.

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